

Roll No.

Total Pages : 3

CMCS/M-22

24543

BIG DATA AND PATTERN RECOGNITION

Paper–MS-20-43

Option–(ii)

Time Allowed : 3 Hours]

[Maximum Marks : 75

Note : Attempt **five** questions in all, selecting **one** question from each Unit. Question No. **1** is compulsory. All questions carry equal marks.

Compulsory Question

1. (a) Define Data? Differentiate between structured, unstructured and semi-structured data.
- (b) How ETL transform multiple source data into single target system? Discuss.
- (c) List the five key concepts that data practitioner and business process owner kept in mind.
- (d) What are the variables to be considered while framing big data environment.
- (e) What is meant by inductive learning?
- (f) Differentiate between supervised and unsupervised pattern recognition.
- (g) List name of NoSQL databases based on : Document, Column and Graph.

- (h) What NoSQL can and cannot do? 15

UNIT–I

2. (a) Elaborate ICT development that have accelerate the pace of big data adoption in business. 8
- (b) What is traditional data visualization? Explain the common features of visualization tools used in Big data. 7
3. What is big data? Discuss its characteristics. Draw a neat diagram and explain nine important stages of the big data analytical life cycle. 15

UNIT–II

4. (a) How to identify the critical dimensions of data quality to manage consumer data expectations? 8
- (b) Write a note on HDFS, Zookeeper, HIVE and Mahout. 7
5. (a) What is big data failure? Discuss the precautions that can salvage some piece of failed big data projects. 8
- (b) Briefly explain the legalities involved for protection of individuals in a big data project. 7

UNIT–III

6. What are Pattern Recognition Systems? Explain the components of a typical pattern recognition system and briefly discuss the potential applications of pattern recognition. 15
7. (a) What is feature extraction and reduction? Discuss different data domains for pattern recognition. 8

- (b) How do you can classify patterns and obtain rules from decision tree? Give a specimen example. 7

UNIT-IV

8. Differentiate between NoSQL and RDBMS. Explain different storage types in NoSQL along with its advantages and disadvantages. 15
9. Write notes on the following :
- (a) CAP theorem.
- (b) NoSQL misconceptions. 15